

Chapter 92

3D Texture Nodes

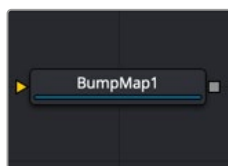
This chapter details the 3D Texture nodes available when creating 3D composites in Fusion. The abbreviations next to each node name can be used in the Select Tool dialog when searching for tools and in scripting references.

For purposes of this document, node trees showing MediaIn nodes in DaVinci Resolve are interchangeable with Loader nodes in Fusion Studio, unless otherwise noted.

Contents

Bump Map [3Bu].....	2034
Catcher [3Ca].....	2037
CubeMap [3Cu].....	2039
Falloff [3Fa].....	2042
Fast Noise Texture [3FN].....	2044
Gradient 3D [3Gd].....	2047
Sphere Map [3SpM].....	2050
Texture 2D [3Tx].....	2052
Texture Transform [3TT].....	2054
The Common Controls	2057

Bump Map [3Bu]



The Bump Map node

Bump Map Node Overview

The Bump Map node is used to convert a grayscale (height map) image into a bump map and takes an input directly from a bump map created by the Create Bump Map node. The node outputs a material.

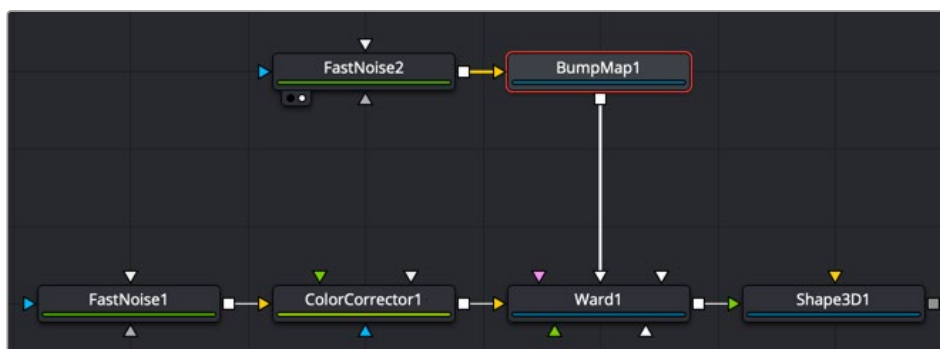
Inputs

The Bump Map node includes a single orange input for connecting a 2D image you want to use as the bump map texture, or it can accept the output of the Create Bump Map node.

ImageInput: The orange Image input is used to connect a 2D RGBA image for the bump calculation or an existing bump map from the Create Bump map node.

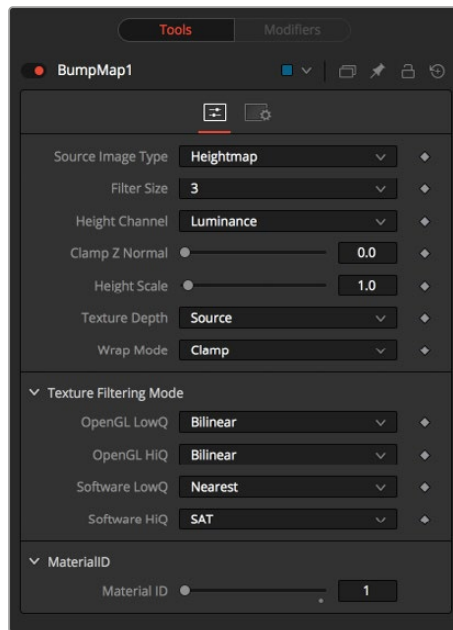
Basic Node Setup

The Bump Map node is connected to the Bump Map material input on any one of the material shader nodes. Below, the example uses a Fast Noise node to generate an image that connects to the Bump Map node. The output of the Bump Map node connects to the Bump Map material input on a Ward node.



A Bump Map is connected to the Bump Map material input on a material node.

Inspector



Bump Map controls

Controls Tab

The Controls tab contains all parameters for modifying the input source and the appearance of the bump map.

Source Image Type

Toggle between Height Map, which creates a bump map similar to the Create Bump Map node, and Bump Map, which expects a bump map created by the Create Bump Map node.

Filter Size

A custom filter generates the bump information. The drop-down menu sets the filter size.

Height Channel

Sets the channel from which to extract the grayscale information.

Clamp Z Normal

Clips the lower values of the blue channel in the resulting bump texture.

Height Scale

Changes the contrast of the resulting values in the bump map. Increasing this value yields a more visible bump map.

Texture Depth

Optionally converts the resulting bump map texture into the desired bit depth.

Wrap Mode

Wraps the image at the borders, so the filter produces correct result when using seamless tile textures.

Common Controls

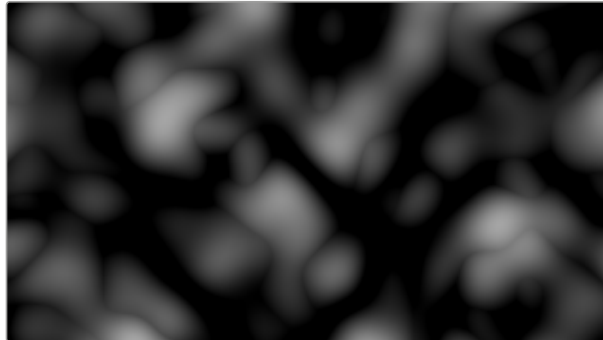
Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Notes on Bump Maps

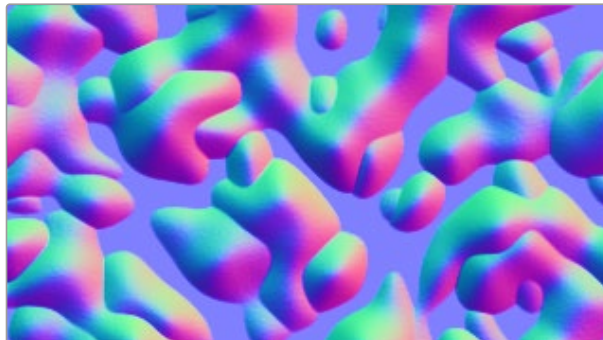
There is some confusion of terminology with bump mapping, depending on where you get your information. Here are Fusion conventions:

Height Map



A grayscale image containing a height value per pixel

Bump Map



An image containing normals stored in the RGB channels used for modifying the existing normals (usually given in tangent space)

Normals Map



An image containing normals stored in the RGB channels used for replacing the existing normals (usually given in tangent or object space)

Catcher [3Ca]



The Catcher node

Catcher Node Overview

The Catcher material is used to “catch” texture-mode projections cast from Projector 3D and Camera 3D nodes. The intercepted projections are converted into a texture map and applied by the Catcher material to the geometry to which it is connected.

To understand the Catcher node, it helps to understand the difference between light-based projections and texture-based projections. Choosing Light from the projection mode menu on the Projector 3D or Camera 3D nodes simply adds the values of the RGB channels in the projected image to the diffuse texture of any geometry that lies within the projection cone. This makes it impossible to clip away geometry based on the Alpha channel of an image when using light mode projections.

Imagine a scenario where you want to project an image of a building onto an image plane as part of a set extension shot. You first rotoscope the image to mask out the windows. This makes it possible to see the geometry of the rooms behind the wall in the final composite. When this image is projected as light, the Alpha channel is ignored, so the masked windows remain opaque.

By connecting the Catcher to the diffuse texture map of the material applied to the image plane, and then switching the projection mode menu in the Projector 3D or Camera 3D node from Light or Ambient Light mode to Texture mode, the projected image is applied as a texture map. When using this technique for the example above, the windows would become transparent, and it would be possible to see the geometry behind the window.

The main advantages of this approach over light projection are that the Catcher can be used to project Alpha onto an object, and it doesn't require lighting to be enabled. Another advantage is that the Catcher is not restricted to the diffuse input of a material, making it possible to project specular intensity maps, or even reflection and refraction maps.

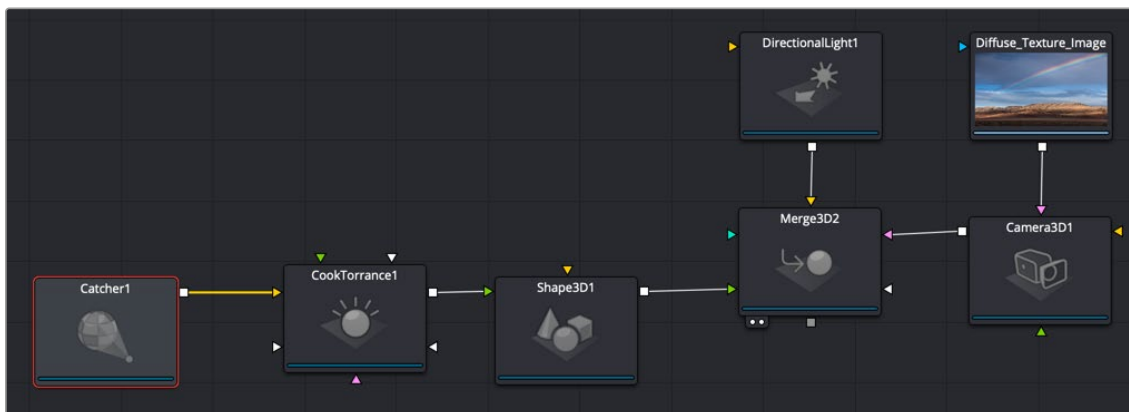
NOTE: The Catcher material requires a Projector 3D or Camera 3D node in the scene, set to project an image in Texture mode on the object to which the Catcher is connected. Without a projection, or if the projection is not set to Texture mode, the Catcher simply makes the object transparent and invisible.

Inputs

The Catcher node has no inputs. The output of the node is connected to the diffuse color material input of the Blinn, Cook Torrance, or other material node applied to the 3D geometry.

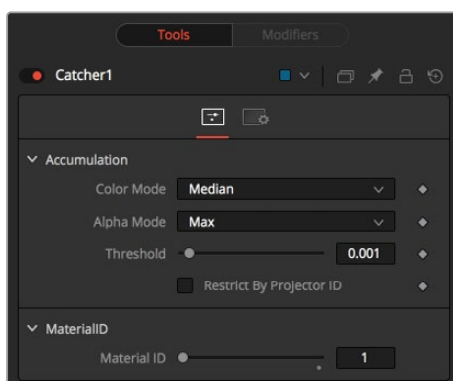
Basic Node Setup

The output of a Catcher node should be connected to the material input of your 3D geometry node. A camera is set up as a projector with an image connected to the camera's image input. When the camera is set to texture projection mode, the Catcher node is used to determine which geometry receives the texture.



A Catcher node output is connected to the input of the geometry node that receives the texture projection

Inspector



Catcher controls

Controls Tab

The Options in the Controls tab determine how the Catcher handles the accumulation of multiple projections.

Enable

Use this checkbox to enable or disable the node. This is not the same as the red switch in the upper-left corner of the Inspector. The red switch disables the tool altogether and passes the image on without any modification. The Enable checkbox is limited to the effect part of the tool. Other parts, like scripts in the Settings tab, still process as normal.

Color Mode

The Color mode menu is used to control how the Catcher combines the light from multiple projectors. It has no effect on the results when only one projector is in the scene. This control is designed to work with the software renderer in the Renderer 3D node and has no effect when using the OpenGL renderer.

Alpha Mode

The Alpha mode is used to control how the Catcher combines the Alpha channels from multiple projectors. It has no effect on the results when only one projector is in the scene. This control is designed to work with the software renderer in the Renderer 3D node and has no effect when using the OpenGL renderer.

Threshold

The Threshold can be used to exclude certain low values from the accumulation calculation. For example, when using the Median Accumulation mode, a threshold of 0.01 would exclude any pixel with a value of less than 0.01 from the median calculation.

Restrict by Projector ID

When active, the Catcher only receives light from projectors with a matching ID. Projectors with a different ID are ignored.

Material ID

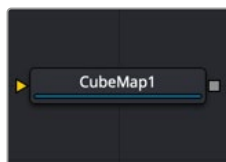
This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the Renderer 3D node.

Common Controls

Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

CubeMap [3Cu]



The Cube Map node

Cube Map Node Overview

The Cube Map node creates texture maps using separate images for each face of the cube. It can also extract the individual faces of the cube from a single image containing an unfolded cube in the Vertical or Horizontal Cross layouts.

A cube map is produced by mounting six cameras at 90 degrees angle of views to point up, down, left, right, front, and back.

The node provides options to set the reference coordinate system and rotation for the resulting texture map. The Cube Map node is typically used to produce environment maps for distant areas (such as skies or horizons) or reflection and refraction maps.



Sample cube map

Inputs

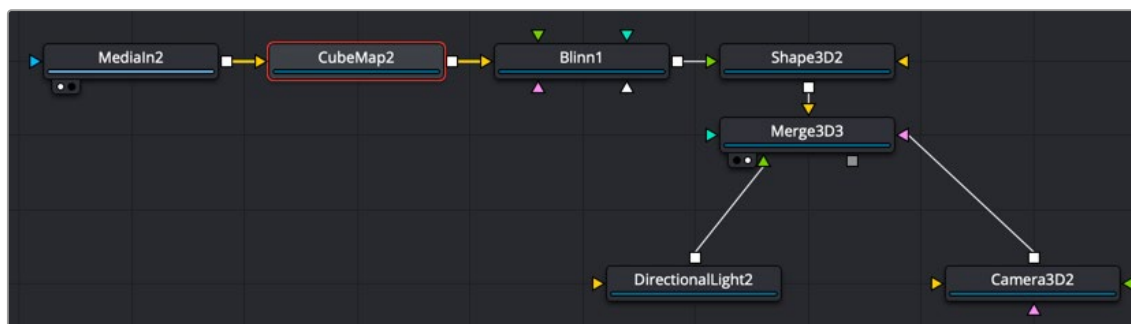
The Inputs on this node change based on the settings of the Layout menu in the Inspector. The single input uses a 2D image for the entire cube, while six inputs can handle a different 2D image for each side of a cube.

CrossImage: The orange Cross Image input is visible by default or when the Layout menu in the Inspector is set to either Vertical Cross or Horizontal Cross. The input accepts a 2D image.

CubeMap.[DIRECTION]: These six multi-colored inputs are visible only when the Layout menu in the Inspector is set to Separate Images. Each input accepts an image aligned to match the left, right, top, bottom, front, and back faces.

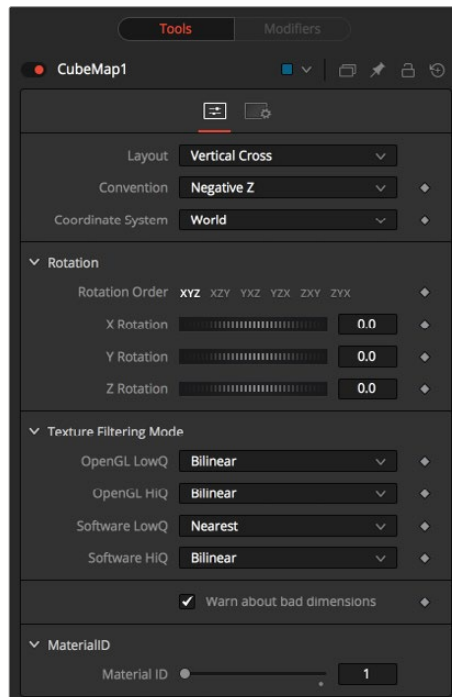
Basic Node Setup

The Cube Map node uses a vertical or horizontal cross image represented by Medialn2 node connected into the orange cross image input. The Cube Map node is used similarly to the Sphere Map node. It creates an environment that surrounds the geometry connected to a Shader node.



A Cube Map node receives a cross image input, creating an environment for the Shape 3D

Inspector



Cube Map controls

Controls Tab

Layout

The Layout menu determines the type and number of inputs for the cube map texture.

Valid options are:

- **Separate Images:** This option exposes six inputs on the node, one for each face of the cube. If the separate images are not square or not of the same size, they are rescaled into the largest 1:1 image that can contain all of them.
- **Vertical Cross:** This option exposes a single input on the node. The image should be an unwrapped texture of a cube containing all the faces organized into a Vertical Cross formation, where the height is larger than the width. If the image aspect of the cross image is not 3:4, the CubeMap node crops it down so it matches the applicable aspect ratio.
- **Horizontal Cross:** This option exposes a single input on the node. The image should be an unwrapped texture of a cube containing all the faces organized into a Horizontal Cross formation, where the width is larger than the height. If the image aspect of the cross image is not 4:3, the CubeMap node crops it down so that matches the applicable aspect ratio.

Coordinate System

The coordinate system menu sets the position values used when converting the image into a texture.

- **Model:** This option orients the texture along the object local coordinate system.
- **World:** This option orients the resulting texture using the global or world coordinate system.
- **Eye:** This option aligns the texture map to the coordinate system of the camera or viewer.

Rotation

The rotation controls are divided into buttons that select the order of rotation along each axis of the texture. For example, XYZ would apply the rotation to the X axis first, followed by the Y axis, and finally the Z axis. The other half of the rotation controls are dials that rotate the texture around its pivot point.

Warn About Bad Dimensions

Selecting this checkbox displays a warning message on the console if the dimensions of the image provided did not meet the requirements of the selected orientation mode.

Material ID

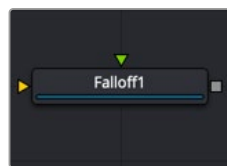
This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

Common Controls

Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

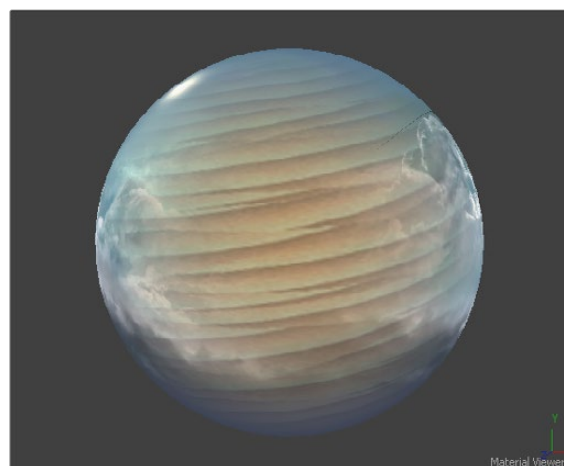
Falloff [3Fa]



The Falloff node

Falloff Node Overview

The Falloff node blends two materials or textures together based on the incidence angle between the object to which the material is applied and the camera. This is useful when you wish to use one material for portions of the geometry that would reflect light directly back to the camera and a different material for parts that reflect light back into the scene.



Falloff example

Inputs

The two Inputs on the Falloff node are used to connect two images or materials. One is used to reflect back at the camera, while the other reflects away from the camera and into the scene.

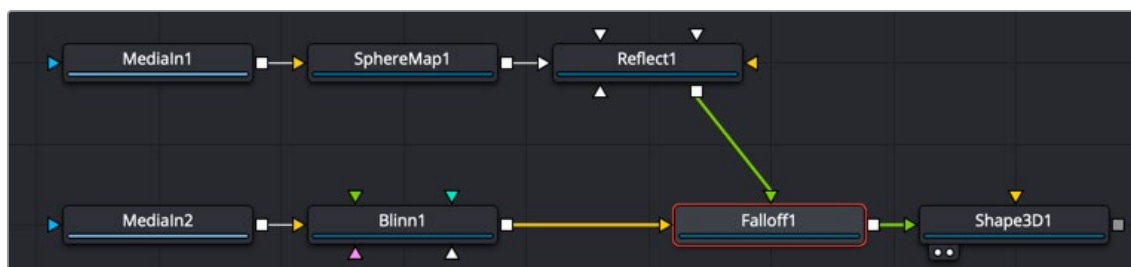
Face On Material: The orange Face On material input accepts a 2D image or a 3D material. If a 2D image is provided, it is turned into a diffuse texture map using the basic material shader. This input is used for the material that is reflecting directly back to the camera

Glancing Material: The green Glancing material input accepts a 2D image or a 3D material. If a 2D image is provided, it is turned into a diffuse texture map using the basic material shader. This input is used for the material that is reflecting away from the camera and into the scene.

While the inputs for this node can be images, the output is always a material.

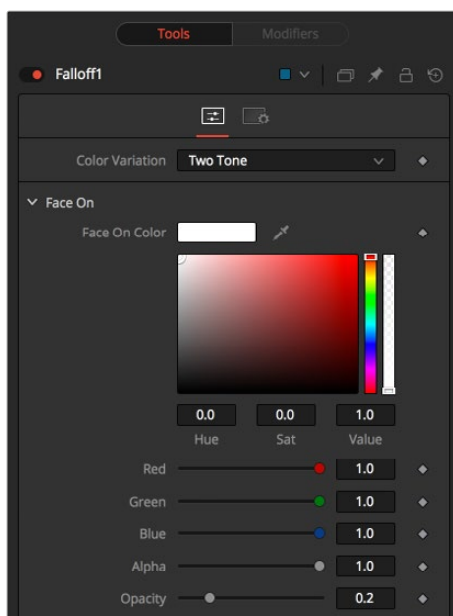
Basic Node Setup

The Falloff node below is used to control the strength of the Blinn material and the Reflect material. You connect the Face On input of the Falloff node to the material you want shown for the sides of the object that face the camera and connect the Glance input to the material you want shown for the sides not directly facing the camera.



The Falloff node uses one input for the material facing the camera and one for the material not directly facing the camera.

Inspector



Falloff controls

Controls Tab

The parameters in the Controls tabs modify the tint and opacity of the Face On material and the Glancing material. A Falloff slider controls the blending between the two.

Color Variation

- **Two Tone:** Two regular Color controls define the colors for Glancing and Face On.
- **Gradient:** A Gradient control defines the colors for Glancing and Face On. This can be used for a multitude of effects, like creating Toon Shaders, for example.

Face On Color

The Face On Color defines the color of surface parts facing the camera. If the Face On texture map is provided, then the color value provided here is multiplied by the color values in the texture.

Reducing the material's opacity decreases the color and Alpha values of the Face On material, making the material transparent.

Glancing Color

The Glancing Color defines the color of surface parts more perpendicular to the camera. If the Glancing material port has a valid input, then this input is multiplied by this color.

Reducing the material's opacity decreases the color and Alpha values of the Glancing material, making the material transparent.

Falloff

This value controls the transition between Glancing and Face On strength. It is very similar to a gamma operation applied to a gradient, blending one value into another.

Material ID

This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

Common Controls

Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in "The Common Controls" section.

Fast Noise Texture [3FN]



The Fast Noise Texture node

Fast Noise Texture Node Overview

The Fast Noise Texture node is the procedural resolution-independent version of the 2D Fast Noise node. It creates a noise texture directly as a material for usage with 3D nodes. It offers a 3D volumetric mode for creating seamless textures in conjunction with nodes providing UVW texture coordinates (Similar to the UV Map 3D node set to XYZ-to-UVW or Camera).

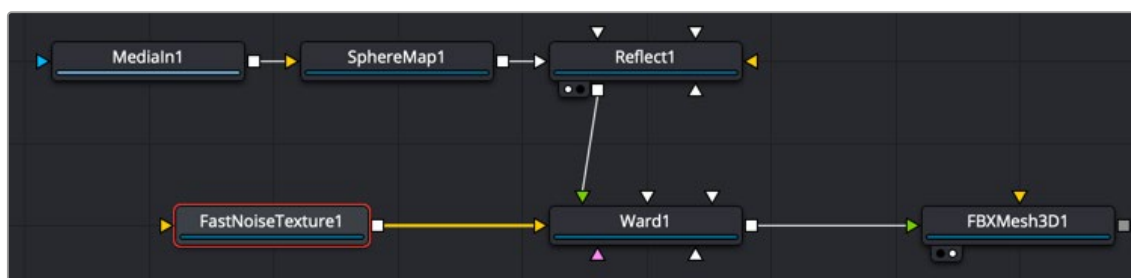
Inputs

The Fast Noise Texture node includes an optional input that can be used to connect a 2D image or material.

SourceMaterial: The Source Materials input accepts a 2D image or a 3D material. The image is then altered by the noise pattern.

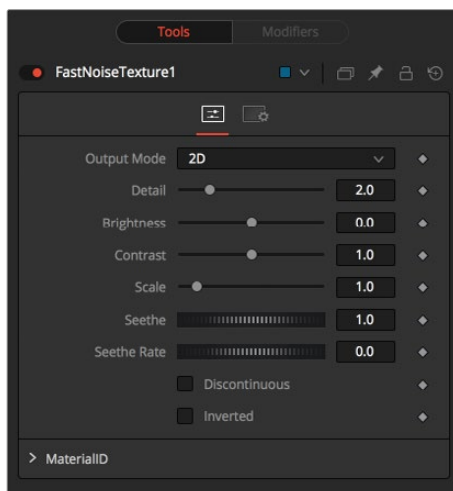
Basic Node Setup

The Fast Noise Texture node below is used to generate a resolution-independent 3D texture for an FBX imported model.



A Fast Noise Texture node generates a seamless texture, taking advantage of UVW coordinates.

Inspector



Fast Noise Texture controls

Controls Tab

The parameters of the Fast Noise Texture node control the appearance and, for 2D, the animation of the noise.

Output Mode

- **2D:** Calculates the noise texture based on 2D texture coordinates (UV). This setting allows smoothly varying the noise pattern with animation.
- **3D:** Calculates the noise texture based on 3D texture coordinates (UVW). Nodes like Shape 3D automatically provide a third texture coordinate; otherwise, a 3D texture space can be created using the UV Map node. The 3D setting does not support animation of the noise pattern.

Detail

Increase the value of this slider to produce a greater level of detail in the noise result. Larger values add more layers of increasingly detailed noise without affecting the overall pattern. High values take longer to render but can produce a more natural result (not all graphics cards support higher detail levels in hardware).

Brightness

This control adjusts the overall Brightness of the noise map.

Contrast

This control increases or decreases the overall Contrast of the noise map. It can exaggerate the effect of the noise.

Scale

The scale of the noise map can be adjusted using the Scale slider, changing it from gentle variations over the entire image to a tighter overall texture effect. This value represents the scale along the UV axis.

Scale Z

(3D only) The Scale Z value scales the noise texture along the W-axis in texture space. W represents a direction perpendicular to the UV plane for a 3D texture map.

Seethe

(2D only) The Seethe control smoothly varies the 2D noise pattern.

Seethe Rate

(2D only) As with the Seethe control above, the Seethe Rate also causes the noise map to evolve and change. The Seethe Rate defines the rate at which the noise changes each frame, causing an animated drift in the noise automatically, without the need for spline animation.

Discontinuous

Normally, the noise function interpolates between values to create a smooth continuous gradient of results. You can enable the Discontinuous checkbox to create hard discontinuity lines along some of the noise contours. The result is a dramatically different effect.

Invert

Enable the Invert checkbox to invert the noise, creating a negative image of the original pattern. This is most effective when Discontinuous is also enabled.

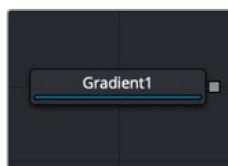
Material ID

This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

Common Controls**Settings Tab**

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Gradient 3D [3Gd]



The Gradient node

Gradient Node Overview

The Gradient 3D node is used to texture objects with a variety of gradient types. It offers many of the same controls as the Background node. While it is not possible to transform the gradient directly in 3D space, it is orientable using the following nodes:

Texture Transform Node: The Texture Transform node can be used to adjust the mapping per pixel.

UV Map Node: The UV Map node can be used to adjust the mapping per vertex (use the XYZtoUVW mode). This has onscreen controls, so you can see what the gradient is doing. Using this node is recommended because it is faster to evaluate.

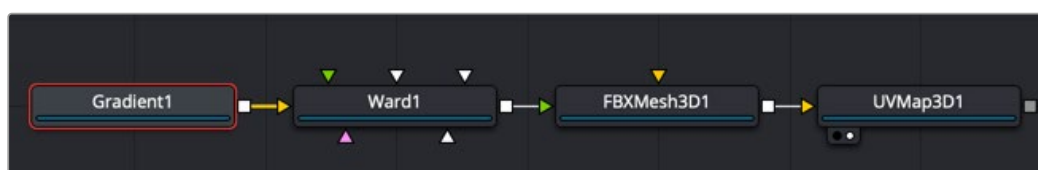
The gradient defaults to a linear gradient that goes from -1 to +1 along the Z-axis. All primitives in the Shape 3D node can output a third texture coordinate for UVW mapping.

Inputs

The Gradient node has no Inputs. The output of the node is connected to a material input on 3D geometry.

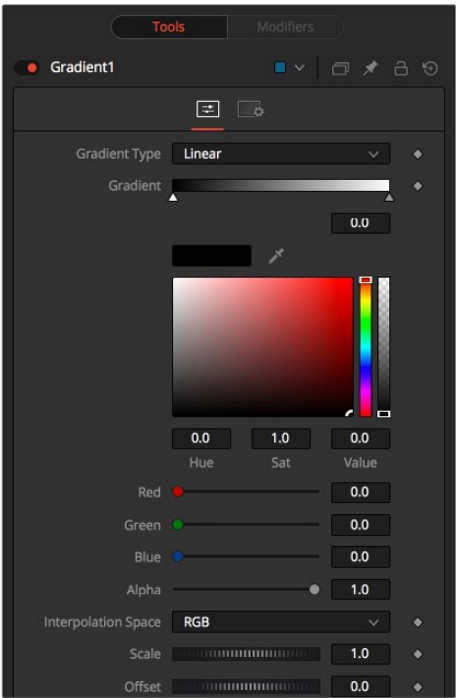
Basic Node Setup

The Gradient 3D node below is used to generate a resolution-independent 3D texture for an FBX imported model. Positioning in UVW space is easiest to do using a UV Map tool placed after the geometry.



A Gradient 3D node generates a resolution-independent gradient texture positioned by the UV Map tool

Inspector



Gradient 3D controls

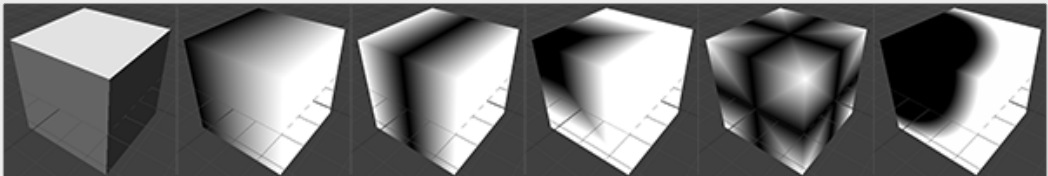
Controls Tab

The Controls tab for the Gradient node control the pattern and colors used for the gradient texture.

Gradient Type

Determines the type or pattern used for the gradient.

- Linear:** A simple linear gradient.
- Reflect:** Based on the Linear mode, this gradient is mirrored at the middle of the textured range.
- Square:** The gradient is applied using a square pattern.
- Cross:** Similar to the Reflect mode, but Cross uses two axes to apply the gradient.
- Radial:** The Radial mode uses a circular pattern to apply the gradient.



Gradient 3D modes

Gradient Bar

The Gradient control consists of a bar where it is possible to add, modify, and remove color stops of the gradient. Each triangular color stop on the Gradient bar represents a color in the gradient. It is possible to animate the color as well as the position of the point. Furthermore, a From Image modifier can be applied to the gradient to evaluate it from an image.

Interpolation Space

The gradient is linearly interpolated from point to point in RGB color space by default. This can sometimes lead to unwanted colors. Choosing another color space may provide a better result.

Scale

Allows sizing of the gradient.

Offset

Allows panning through the gradient.

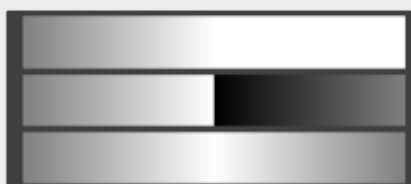
Repeat

Defines how the left and right borders of the gradient are treated.

Once: When using the Gradient Offset control to shift the gradient, the border colors keep their values. Shifting the default gradient to the left results in a white border on the left, while shifting it to the right results in a black border on the right.

Repeat: When using the Gradient Offset control to shift the gradient, the border colors wrap around. Shifting the default gradient to the left results in a sharp jump from white to black, while shifting it to the right results in a sharp jump from black to white.

Ping Pong: When using the Gradient Offset control to shift the gradient, the border colors ping-pong back and forth. Shifting the default gradient to the left results in the edge fading from white back to black, while shifting it to the right results in the edge fading from black back to white.



Gradients set to Once, Repeat, and Ping Pong from top to bottom, respectively, and shifting the gradient to the left

Sub Pixel

Determines the accuracy with which the gradient is created.

Material ID

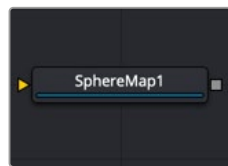
This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

Common Controls

Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Sphere Map [3SpM]



The SphereMap node

Sphere Map Node Overview

The Sphere Map node can be used to create simulated environment mapping, also called reflection mapping. Ray trace rendering a reflective scene can be very time consuming, but sphere map-based reflection mapping can generate 360-degree reflections faster with little loss of accuracy. For example, when creating a reflective environment, a sphere map is created, large enough to surround the 3D object in your scene. The sphere is mapped with the environment you want reflected and connected to the Reflection Color input on a Reflect node.

Inputs

The single image input on the Sphere Map node accepts a 2D image texture in an equirectangular format (where the X-axis represents 0–360 degrees longitude, and the Y-axis represents –90 to +90 degrees latitude.)

ImageInput: The orange Image input accepts a 2D RGBA image. Preferably, this is an equirectangular image that shows the entire vertical and horizontal angle of view up to 360 degrees.

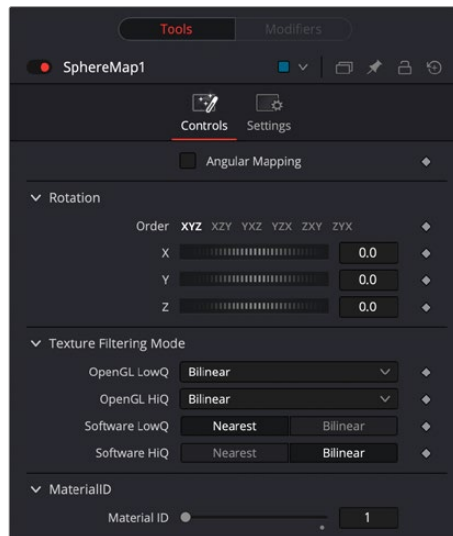
Basic Node Setup

The Sphere Map node below is mapped with a spherical image to generate the environment reflected on the Shape 3D. It is connected to the Reflection Color input on a Reflect node.



A Sphere Map node generates a reflective environment when connected to a Reflect node Reflection Color input.

Inspector



Sphere Map controls

Controls Tab

The Controls tab in the Inspector modifies the mapping of the image input to the sphere map.

Angular Mapping

Adjusts the texture coordinate mapping so the poles are less squashed and areas in the texture get mapped to equal areas on the sphere. It turns the mapping of the latitude lines from a hemispherical fisheye to an angular fisheye. This mapping attempts to preserve area and makes it easier to paint on or modify a sphere map since the image is not as compressed at the poles.

Rotation

Offers controls to rotate the texture map.

Material ID

This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

The node expects an image with an aspect ratio of 2:1. Otherwise, the image is clamped according to the following rules:

- **2 * width > height:** The width is fitted onto the sphere, and the poles display clamped edges.
- **2 * width < height:** The height is fitted onto the sphere, and there is clamping about the 0-degree longitude line.

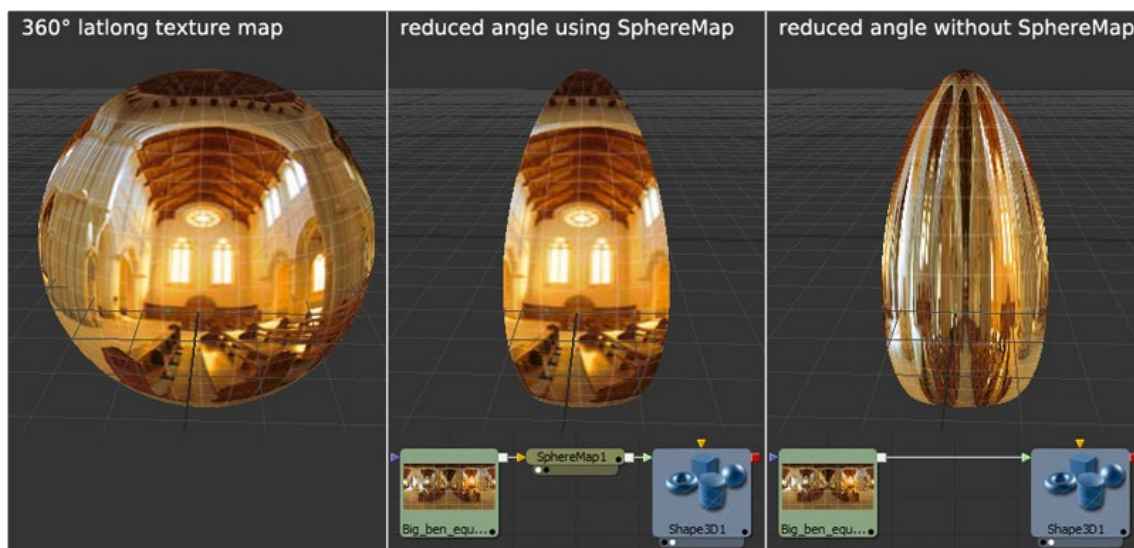
Common Controls

Settings tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Sphere Map vs. Connecting the Texture to a Sphere Directly

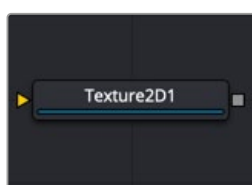
You can connect an equirectangular texture map directly to a sphere instead of piping it through the Sphere Map node first. This results in a different rendering if you set the start/end angle and latitude to less than 360°/180°. In the first case, the texture is squashed. When using the Sphere Map node, the texture is cropped. Compare:



Spherical mapping differences

NOTE: If you pipe the texture directly into the sphere, it is also mirrored horizontally. You can change this by using a Transform node first.

Texture 2D [3Tx]



The Texture 2D node

Texture Node Overview

The Texture 2D node sets metadata of an image being used for a texture map. By default, an image will be (0,0) to (1,1) UV, but that can be changed. The Texture node relies on the presence of U and V Map channels in 3D rendered images. If these channels are not present, this node has no effect.

NOTE: Background pixels may have U and V values of 0.0, which set those pixels to the color of the texture's corner pixel. To restrict texturing to specific objects, use an effect mask based on the Alpha of the object, or its Object or Material ID channel.

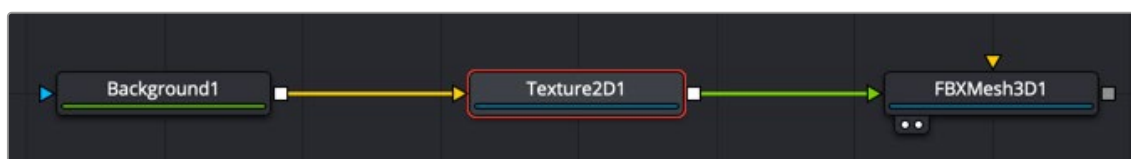
For more information, see *Chapter 78, “Understanding Image Channels,”* in the DaVinci Resolve Reference Manual, or *Chapter 18* in the Fusion Reference Manual.

Inputs

Image Input: The orange image input expects a 2D image.

Basic Node Setup

The Texture 2D node below takes a 2D gradient from the Background node and sets the UV metadata for it. The texture is then applied to the FBX geometry based on that metadata. If you have the option to use the UV Map tool, it is recommended because it may be faster and has onscreen controls.



A Texture 2D node is used to set the 3D texture metadata for the input image.

Inspector



Texture 2D controls

Controls Tab

The Controls tab of the Inspector includes the following options.

U/V Offset

These sliders can be used to offset the texture along the U and V coordinates.

U/V Scale

These sliders can be used to scale the texture along the U and V coordinates.

Wrap Mode

If a texture is transformed in the texture space (using the controls below or the UV Map node), then it's possible that areas beyond the image borders will be mapped on the object. The Wrap Mode determines how the image is applied in these areas.

- **Wrap:** This wraps the edges of the image around the borders of the image.
- **Clamp:** The color at the edges of the images is used for texturing. This mode is similar to the Duplicate mode in the Transform node.
- **Black:** The image is clipped along its edges. A black color with Alpha = 0 is used instead.
- **Mirror:** The image is mirrored in both X and Y.

Texture Filtering Mode

The texture can be filtered differently depending on whether you are using the Software Renderer or OpenGL renderer in the Renderer 3D node. Within the two render engines, you can choose between high-quality anti-aliasing or low quality. The texture filtering mode provides different filtering options for the two render engines and the two anti-aliasing settings.

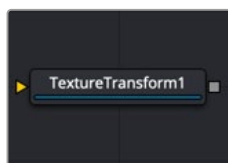
- **Nearest:** The simplest filtering technique is very fast but can cause artifacts when scaling textures.
- **Bilinear:** A standard isotropic filtering technique for scaling textures into multiple resolutions. Works well for magnification of textures.
- **Trilinear:** An extension of Bilinear filtering. Trilinear tends to be a better option when scaling down textures
- **Anisotropic:** The highest-quality filtering method that takes the camera orientation and polygon perspective into account.
- **SAT:** SAT (Summed Area Table) is a method of performing high-quality filtering, but it can require more memory than other options. Works very well on smaller bitmaps.

Common Controls

Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in detail at the end of this chapter in “The Common Controls” section.

Texture Transform [3TT]



The Texture Transform node

Texture Transform Node Overview

The Texture Transform node can be used to translate, rotate, and scale the UVW texture coordinates of a 3D object. While the input can also be an image, the output is always a material.

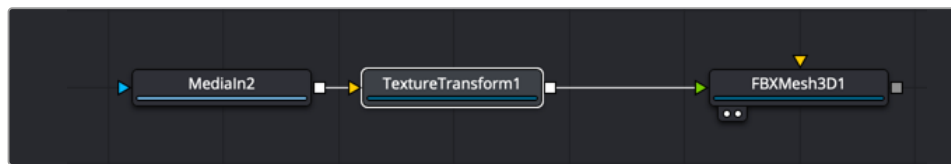
Inputs

The Texture Transform node includes a single input that is used to connect the image or material you want to transform.

Material Input: The orange Material input accepts a 2D image or 3D material whose texture coordinates are transformed using the controls in the Inspector.

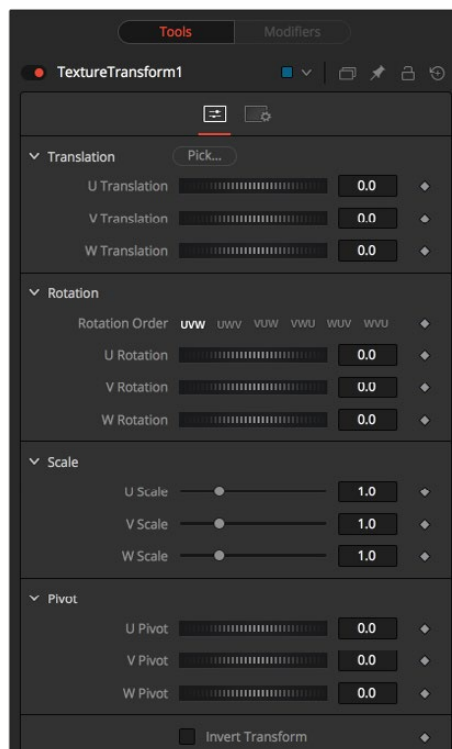
Basic Node Setup

The Texture Transform node below is used to take in a 2D image, transform it, and output a material to be used on 3D geometry.



A Texture Transform node transforms a texture applied to 3D geometry.

Inspector



Texture Transform controls

NOTE: Not all Wrap modes are supported by all graphics cards.

Controls Tab

The Controls tab for the Texture Transform node includes many common transform controls that are used to transform the texture using UVW coordinates.

Translation

The U, V, W translation sliders shift the texture along U, V, and W axes.

Rotation

Rotation Order buttons set the order in which the rotation is applied. In conjunction with the buttons, the UVW dials define the rotation around the UVW axes.

Scale

U, V, W sliders scale the texture along the UVW axes.

Pivot

U, V, W Pivot sets the reference point for rotation and scaling.

Material ID

This slider sets the numeric identifier assigned to this material. This value is rendered into the MatID auxiliary channel if the corresponding option is enabled in the renderer.

Common Controls

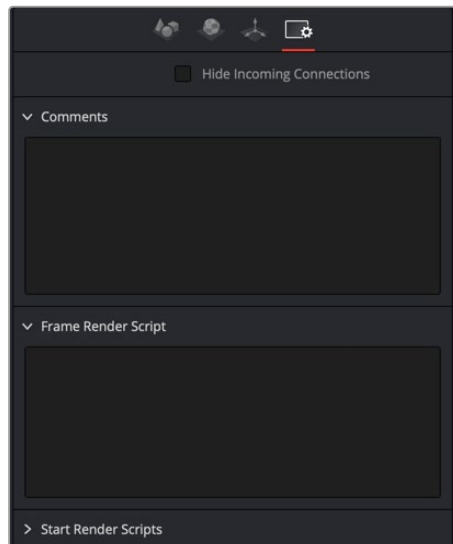
Settings Tab

The Settings tab in the Inspector is duplicated in other 3D nodes. These common controls are described in the following “The Common Controls” section.

The Common Controls

Nodes that handle 3D geometry share a number of identical controls in the Inspector. This section describes controls that are common among 3D Texture nodes.

Settings Tab



Common Settings 3D controls

The Common Settings tab can be found on most tools in Fusion. The following controls are specific settings for 3D nodes.

Hide Incoming Connections

Enabling this checkbox can hide connection lines from incoming nodes, making a node tree appear cleaner and easier to read. When enabled, fields for each input on a node are displayed. Dragging a connected node from the node tree into the field hides that incoming connection line as long as the node is not selected in the node tree. When the node is selected in the node tree, the line reappears.

Comment Tab

The Comment tab contains a single text control that is used to add comments and notes to the tool. When a note is added to a tool, a small red dot icon appears next to the setting's tab icon, and a text bubble appears on the node. To see the note in the Node Editor, hold the mouse pointer over the node for a moment. The contents of the Comments tab can be animated over time, if required.

Scripting Tab

The Scripting tab is present on every tool in Fusion. It contains several edit boxes used to add scripts that process when the tool is rendering. For more details on the contents of this tab, please consult the scripting documentation.